

## **From molecule to reads: sequencing, coverage and quality**

- BIB — Bioinformatics for Bioengineering / Biyomühendislik için Biyoinformatik
- Week 5 · Block B "Is it real?"
- Molekülden okumalara: dizileme, kapsam ve kalite
- Today the archetype assignment runs for the first time: Design Notebook 1

## **Ad of the Week #5**

- Skills Ledger tick: "platform choice and sequencing QC"

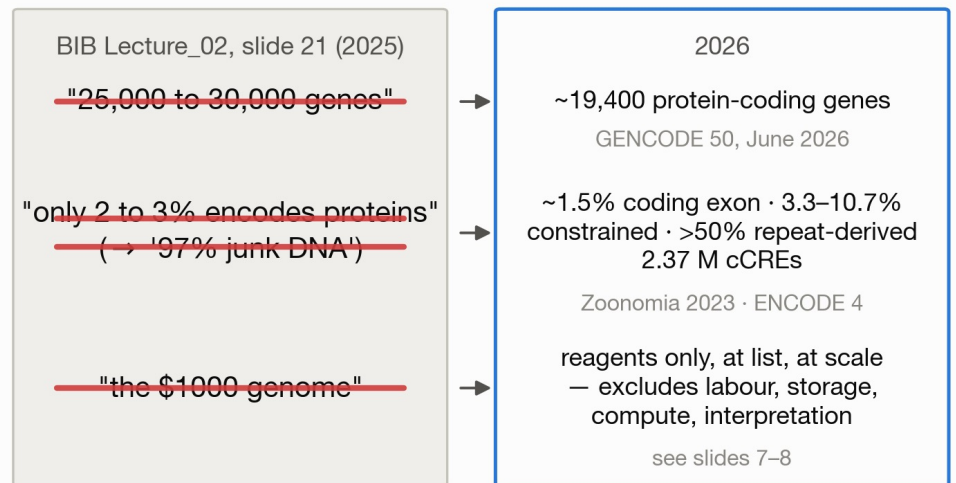
# Sequencing history in four minutes

- Sanger 1977: chain termination, <1 kb reads, one molecule at a time
- Shotgun: break the genome into random pieces, sequence, reassemble
- The Human Genome Project ran 1990–2003, ~13 years, ~\$2.7 bn. That is the whole story.
- Three numbers on last year's slides are wrong. They are retired on the next slide.

# Three numbers we are retiring today

- RETIRED: "25,000–30,000 genes" → ~19,400 protein-coding genes (GENCODE 50, June 2026)
  - Catalogues disagree on ~1 in 8 borderline genes (Maquedano et al., Database 2025)
- RETIRED: "97% junk DNA" → ~1.5% coding exon; ~3.3–10.7% evolutionarily constrained (Zoonomia, Science 2023)
  - Also >50% repeat-derived, and 2.37 million candidate cis-regulatory elements in ENCODE 4
- RETIRED: "the \$1000 genome" → see the next two slides

ERRATA — three numbers from last year's deck, retired today



'Junk' is not a defensible category; 'unconstrained' is — and the share depends on method.

*Errata: the 2025 numbers struck through, the 2026 numbers with sources*

# The 2026 platform decision table — short reads

| Platform                | Chemistry         | Cost / 30× genome         | Read length        | Who runs it                |
|-------------------------|-------------------|---------------------------|--------------------|----------------------------|
| NovaSeq X Plus (25B)    | SBS               | ~\$200 list, consumables  | 2×150              | most cores; WGS, RNA-seq   |
| Element AVITI / AVITI24 | avidity           | ~\$100 claimed            | 2×150–300          | mid-scale cores, multiomic |
| Ultima UG100 / UG200    | wafer, single-end | ~\$80                     | ~300 nt single-end | mega-cohorts, single-cell  |
| MGI DNBSEQ-T20×2        | DNB / cPAS        | ~\$1/Gb, sub-\$100 all-in | 2×150              | national genome programmes |

# Short reads: what the table does not say

- These are list reagent costs at scale. They are not what an experiment costs.
- NovaSeq X Plus remains the default for population WGS/WES at most cores
- Ultima also targets cfDNA and single-cell at scale; AVITI, multiomic readouts
- MGI's sub-\$100 figure includes instrument depreciation, and suits cost-sensitive labs
- Ultima's UG200 launched at AGBT on 25 Feb 2026 and shipped Q2 2026.

Short-read platforms, 2026 — list reagent cost at scale, not what an answer costs

| Platform                      | Chemistry         | Cost / 30× genome                     | Read length | Who runs it                                |
|-------------------------------|-------------------|---------------------------------------|-------------|--------------------------------------------|
| Illumina NovaSeq X Plus (25B) | SBS               | ~\$200 list, consumables only         | 2×150       | population WGS/WES, RNA-seq, most cores    |
| Element AVITI / AVITI24       | avidity           | ~\$100 claimed                        | 2×150–300   | mid-scale cores, multiomic readouts        |
| Ultima UG100 / UG200          | wafer, single-end | ~\$80                                 | ~300 nt SE  | mega-cohorts, single-cell at scale, cfDNA  |
| MGI DNBSEQ-T20×2              | DNB / cPAS        | sub-\$100 incl. depreciation, ~\$1/Gb | 2×150       | national genome programmes, cost-sensitive |

*Short-read platforms, 2026 — list reagent cost at scale*

# The 2026 platform decision table — long reads

| Platform                 | Chemistry        | Cost                      | Accuracy             | Read length       |
|--------------------------|------------------|---------------------------|----------------------|-------------------|
| PacBio Revio + SPRQ      | HiFi SMRT        | \$345 list                | HiFi Q30–Q40         | 15–25 kb          |
| PacBio Vega              | benchtop HiFi    | —                         | HiFi                 | 15–25 kb          |
| ONT PromethION / GridION | nanopore R10.4.1 | per flow cell, not genome | duplex ~99.92%, ~Q30 | 10 kb – megabases |

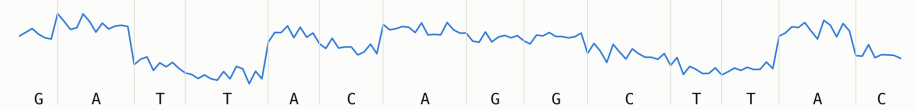
# Long reads: what the table does not say

- Both call 5mC and 5hmC natively, in the same run, no bisulfite — per molecule, phased
- ONT direct RNA (SQK-RNA004): native RNA, poly(A) length, m6A, full-length isoforms, no RT
- PacBio Revio: \$345 list, sub-\$300 at  $\geq 5,000$  genomes/yr
- ONT R10.4.1: simplex  $\sim 99.6\%$  (SUP), duplex  $\sim 99.92\%$ ,  $\sim Q30$
- Adaptive sampling — the software rejects unwanted molecules mid-read
  - Honest number:  $\sim 1.9\times$  enrichment for direct RNA, far below hybridisation capture
  - It wins on turnaround, not depth

Long-read platforms, 2026 — accuracy is no longer a short-read advantage

| Platform                          | Chemistry     | Cost                                      | Accuracy                                                          | Read length       |
|-----------------------------------|---------------|-------------------------------------------|-------------------------------------------------------------------|-------------------|
| PacBio Revio + SPRQ               | HiFi SMRT     | \$345 list; sub-\$300 at $\geq 5,000$ /yr | HiFi Q30–Q40                                                      | 15–25 kb          |
| PacBio Vega                       | benchtop HiFi | —                                         | HiFi                                                              | 15–25 kb          |
| ONT PromethION / GridION, R10.4.1 | nanopore      | per flow cell, not per genome             | simplex $\sim 99.6\%$ (SUP)<br>duplex $\sim 99.92\%$ , $\sim Q30$ | 10 kb – megabases |

raw ionic current (pA, simulated squiggle) → basecalled letters underneath — see slide 13



Long-read platforms, 2026, with a simulated nanopore squiggle inset

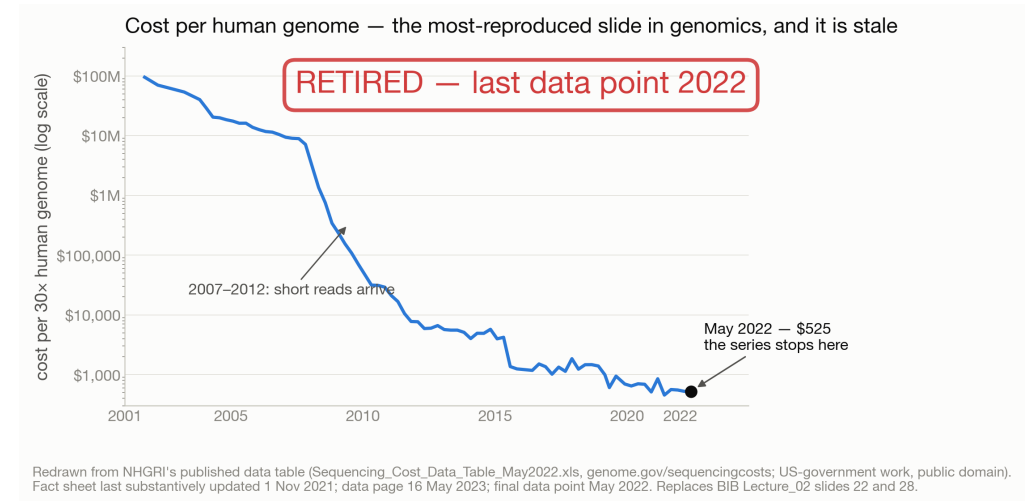


# Three teaching points that outlive the table

- 1. Accuracy is no longer a short-read advantage
  - The trade is now cost and throughput versus read length and native base modifications
- 2. Methylation is no longer a separate experiment on long reads
  - Short reads: EM-seq (enzymatic, TET2+APOBEC) replaced bisulfite
  - ~54M CpGs covered vs ~45M at 200 ng input; better GC uniformity, no heat/pH degradation
- 3. Illumina is no longer synonymous with sequencing
  - A graduate joining a Turkish or EU core lab in 2026 may meet AVITI, DNBSEQ or Revio first
- Trichotomy check (Week 4's table): which row does a basecaller sit in? Hold that for ten minutes.

# The cost curve is dead. Retire it.

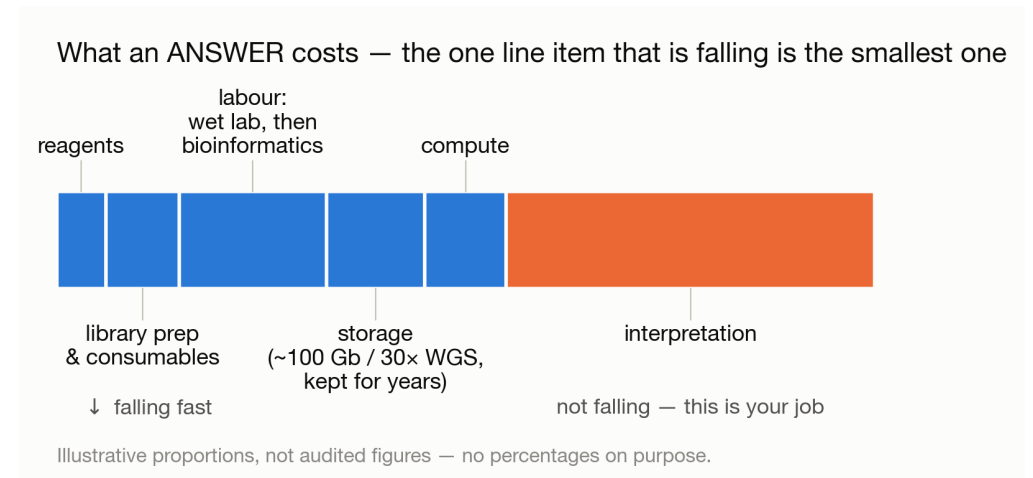
- The NHGRI cost-per-genome fact sheet was last substantively updated 1 November 2021; the data page 16 May 2023; the last data point is 2022
- Quoting it in 2026 is an error of about four years
- It stays in this deck as one image, captioned RETIRED, because you will meet it in other people's talks
- And because "is this number still true?" is the skill, not "what is the number"



*RETIRED — NHGRI cost per genome, series ends May 2022 at \$525*

# What an answer costs

- "Cost per genome" = reagents, on the instrument, at list price, at scale. It excludes almost everything.
- The real cost of an ANSWER, roughly:
  - Reagents — a minority share, and falling fast
  - Library preparation and consumables
  - Labour: wet-lab, then bioinformatics
  - Storage and compute: ~100 Gb of raw data per 30× human WGS, kept for years
  - Interpretation — and this line is not falling
- Interpretation is your future job. That is not a motivational slogan; it is where the money went.



*Illustrative proportions, not audited figures*

# What a read actually is

- Line 1: @ + identifier. Line 2: the bases. Line 3: +. Line 4: one quality character per base.
- Quality is Phred+33: char code minus 33 = Q. ! = Q0, I = Q40.
- $Q = -10 \cdot \log_{10}(P_{\text{error}})$ . So Q30 = 1 error in 1,000 = 99.9% accurate. Q20 = 1 in 100.
- Line 2 and line 4 must be the same length. If they are not, your file is corrupt.
- A paired-end fragment produces TWO records. Read this sentence again. Design Notebook 1 turns on it.

```
@SRR1039512.1 HWI-ST177:314:C12K6ACXX:1:1101:1207:2192 length=63
ACGACAGGTCATCGTTCAAGCAGAATGCAGACAGGCCATTACGAGCCCAAGTTGAAGAGAAG
+SRR1039512.1 HWI-ST177:314:C12K6ACXX:1:1101:1207:2192 length=63
FBDEFIDH2AGHIICDHGBDDFGGGDH<FFEA;FH=;BGGA@DFGB<EEE=??CCC>C>;=A?
```

One real FASTQ record — example\_file.fastq, record 1 (SRR1039512)

```
1 @SRR1039512.1 HWI-ST177:314:C12K6ACXX:1:1101:1207:2192 length=63
  '@' then the identifier: run, instrument, flowcell, lane, tile, x, y

2 ACGACAGGTCATCGTTCAAGCAGAATGCAGACAGGCCATTACGAGCCCAAGTTGAAGAGAAG
  the bases — 63 of them

3 +SRR1039512.1 HWI-ST177:314:C12K6ACXX:1:1101:1207:2192 length=63
  '+' — the identifier may be repeated here or left empty

4 FBDEFIDH2AGHIICDHGBDDFGGGDH<FFEA;FH=;BGGA@DFGB<EEE=??CCC>C>;=A?
  ONE quality character per base — 63 of them. Lengths of lines 2 and 4 must match, or the file is corrupt
```

Phred+33:  $Q = \text{ASCII code} - 33$      $P(\text{error}) = 10^{-(Q/10)}$

!

33 – 33 = Q0  
 $P(\text{error}) = 1$   
no confidence at all

5

53 – 33 = Q20  
 $P(\text{error}) = 0.01$   
error 1 in 100

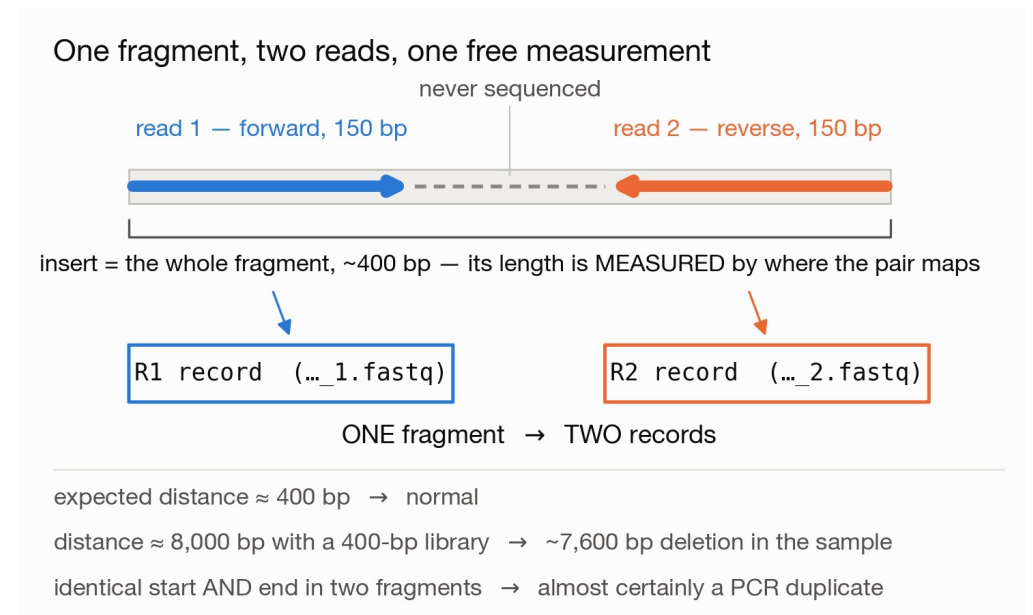
I

73 – 33 = Q40  
 $P(\text{error}) = 0.0001$   
error 1 in 10,000

*Record 1 of example\_file.fastq (SRR1039512), with the Phred+33 strip*

# Insert size, and why the pair is a measurement

- The insert is the whole fragment. The two reads are its two ends. The middle is never sequenced.
- The distance between the mapped pair is therefore a MEASUREMENT of fragment length
- Which is why paired reads detect structural variants: an unexpected insert size means something moved
- And why duplicate marking works: two fragments with identical start AND end are almost certainly PCR copies
- Single-end throws all of this away. Sometimes that is fine (counting), sometimes it is not (structure).



*One fragment, two reads, one free measurement*

# Coverage: $C = N \cdot L / G$

- $C$  = mean coverage (kapsam) ·  $N$  = number of reads ·  $L$  = read length ·  $G$  = genome size
- Worked example 1, single-end. Bacterial genome,  $G = 5$  Mb. Reads  $L = 150$  bp.  $N = 2,000,000$ .
  - $C = (2 \times 10^6 \times 150) / (5 \times 10^6) = 60\times$
- Worked example 2, paired-end. You buy 1 Gb of sequence for the same 5 Mb genome,  $2 \times 150$ .
  - Total bases =  $1 \times 10^9$ , so  $C = 1 \times 10^9 / 5 \times 10^6 = 200\times$
  - The trap:  $1 \times 10^9 / 150 = 6.67\text{M READS} = 3.33\text{M FRAGMENTS}$ . Count fragments  $\times 150$  and you get  $100\times$ . Wrong by exactly 2.
- Human scale, for calibration:  $1\times$  of a 3.1 Gb genome at  $L = 150$  needs  $\sim 20.7$  million reads.  $30\times$  needs  $\sim 620$  million.

$C = N \cdot L / G$  — two worked examples, one trap

1 · single-end

$G = 5$  Mb,  $L = 150$  bp

$N = 2,000,000$  reads

$$C = (2 \times 10^6 \times 150) / (5 \times 10^6) \\ = 3 \times 10^8 / 5 \times 10^6 = 60\times$$

2 · paired-end, 1 Gb bought,  $2 \times 150$

total bases =  $1 \times 10^9$ ,  $G = 5$  Mb

$1 \times 10^9 / 150 = 6.67$  M reads

= 3.33 M fragments

$$C = 1 \times 10^9 / 5 \times 10^6 = 200\times$$

THE TRAP — count fragments as if they were reads

$$C = (3.33 \times 10^6 \times 150) / (5 \times 10^6) = 100\times$$

Wrong by exactly 2 — it runs clean, plots fine, and nothing warns you.

This is the defect planted in this afternoon's audit.

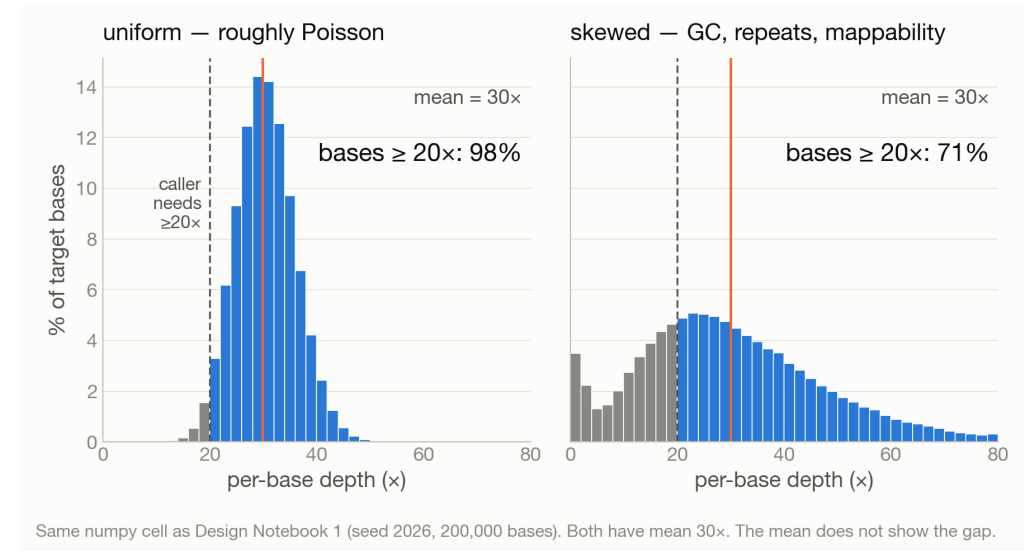
Human calibration,  $G = 3.1$  Gb,  $L = 150$ :  $1\times \rightarrow 3.1 \times 10^9 / 150 \approx 20.7$  M reads  $30\times \rightarrow \approx 620$  M reads

$N$  counts READS or FRAGMENTS — decide, then be consistent. The factor between them is 2.

$C = N \cdot L / G$  — two worked examples, one trap

# But the mean is a lie: uniformity, and what the caller wants

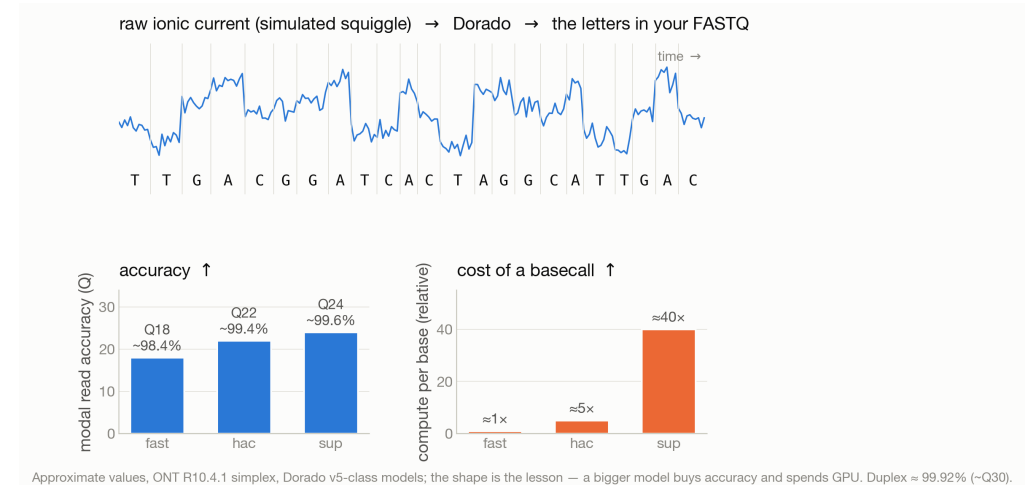
- Reads do not land uniformly. GC-rich and GC-poor regions, repeats, capture bias, and mappability all skew it.
- A 30× mean with a skewed distribution still leaves callable gaps. The mean does not show them.
- So the number that matters is not the mean — it is the fraction of your target at or above the depth your caller needs
- And that threshold is dictated by the downstream tool, not by a formula:
  - Germline SNV calling: conventionally ~30× WGS
  - Somatic variants at low allele fraction: 100× to 1000×+
  - Simply detecting presence/absence: much less
- In this afternoon's notebook you will compute exactly this, on your own seed: the fraction of target at  $\geq 10\times$  and  $\geq 20\times$ .



*Same mean 30×; 98% vs 71% of bases at  $\geq 20\times$*

# A neural network made every base call in your FASTQ

- Nanopore does not read bases. It reads ionic current — a squiggle. Dorado translates the squiggle into letters.
- The fast / hac / sup menu is not a quality setting. It is a model size vs accuracy vs compute trade-off.
  - This is the core machine-learning trade-off, and you just met it with zero mathematics
- Duplex reads both strands of the same molecule through a stereo model: ~99.92%, ~Q30
- Remora calls 5mC, 5hmC and 6mA from the SAME raw signal — methylation with no conversion chemistry, per molecule, phased
- PacBio's equivalent: DeepConsensus, a transformer that polishes subreads into HiFi reads
- Trichotomy check: a basecaller is learned. What was it trained on?



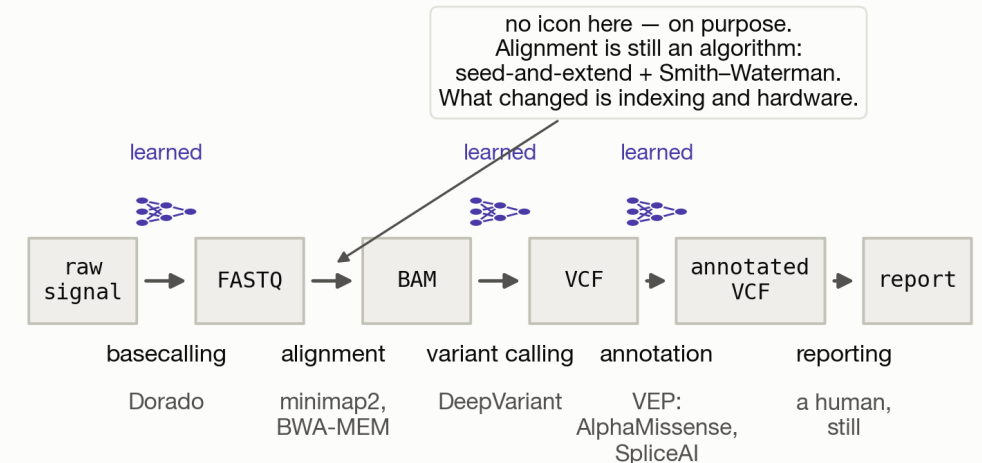
*Simulated squiggle → Dorado → letters; accuracy up, compute up*



# EM-seq, and where the AI sits in your pipeline

- Short-read methylation in 2026 is EM-seq, not bisulfite: enzymatic (TET2 + APOBEC), no heat/pH degradation, ~54M CpGs vs ~45M at 200 ng input
- Where deep learning already sits, whether or not anyone told you:
  - Basecalling — Dorado, DeepConsensus
  - Variant calling — DeepVariant 1.10 (a CNN reading a pileup as an image)
  - Variant interpretation — CADD v1.7 embeds ESM-1v; VEP ships AlphaMissense and SpliceAI plugins
- DeepVariant did not learn biology. It learned to read a pileup image the way a human curator reads IGV.
- None of this is a future topic. It is already inside the files you were given.

Where a neural network already sits in your pipeline



*Learned steps carry a network icon; the alignment arrow deliberately does not*

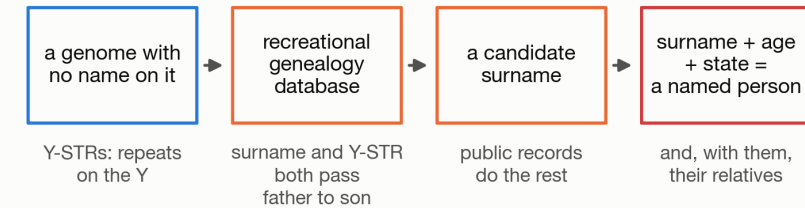
# QC is a decision, not a ritual

- Walk the modules: per-base quality · per-base sequence content · duplication · adapter content · overrepresented sequences · GC distribution
- The framing question, every single time: what would I do differently if this module were green?
  - If the answer is "nothing", the flag is not actionable and you move on
- Some red flags are expected artefacts of the library type, not problems
- 2026 defaults: fastp for trimming and filtering; MultiQC to aggregate. Trimmomatic is retired from every handout in this course.

# Ethics beat: genomic data cannot be anonymised

- Not "is hard to anonymise". Cannot. Your genome identifies you, and partly your relatives.
- Surname inference (Gymrek et al., Science 2013): Y-chromosome STRs plus public recreational genealogy databases
  - Recovered surnames, and then identities, for men in a supposedly de-identified research cohort
- Under KVKK and GDPR, genetic data is special-category personal data (özel nitelikli kişisel veri)

How a "de-identified" genome gets a name on it



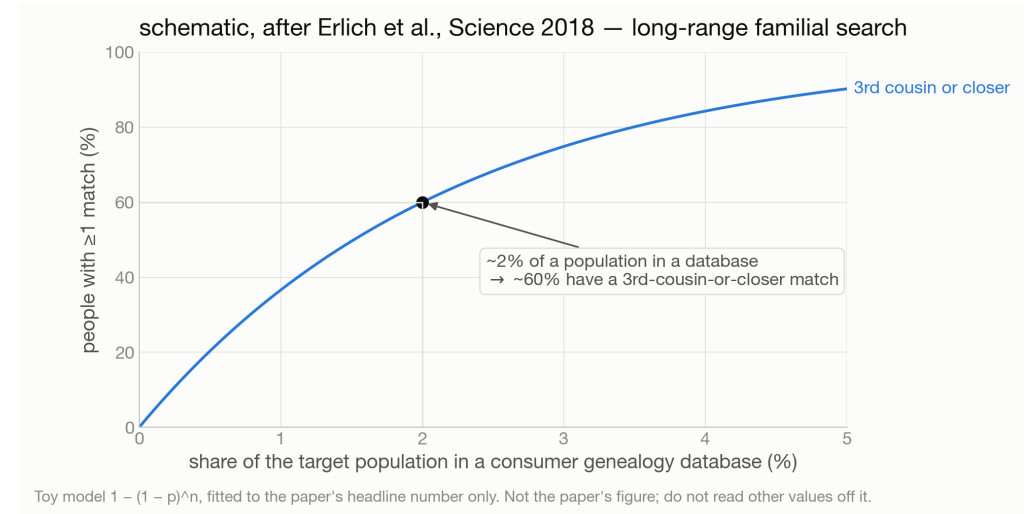
Nothing here is a hack: every step uses a public resource for the purpose it was built for.

Schematic of the method in Gymrek M, McGuire AL, Golan D, Halperin E, Erlich Y, Science 2013;339(6117):321–324 — redrawn, NOT the paper's figure.

*Schematic of the method in Gymrek et al., Science 2013 — redrawn, not the paper's figure*

# Re-identification: the 2018 result and the 2026 twist

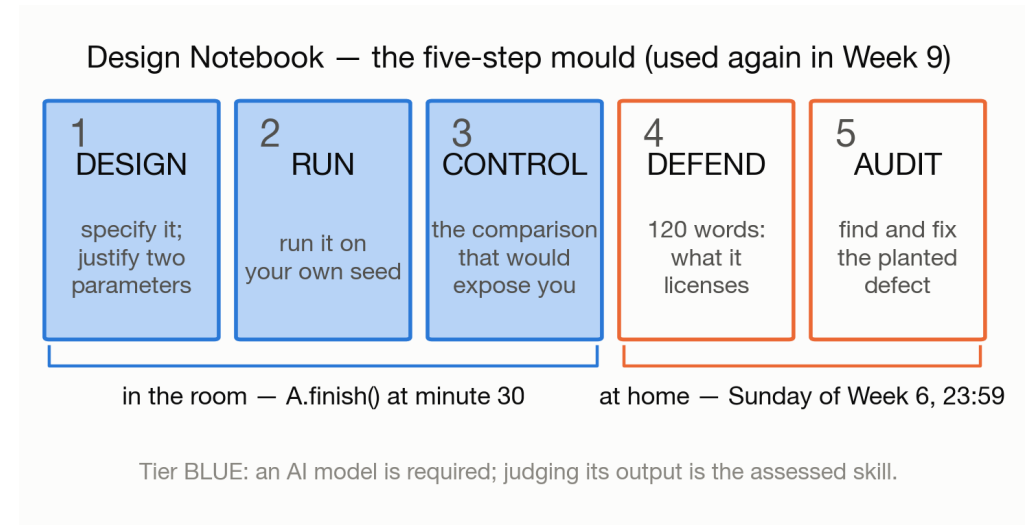
- Long-range familial search (Erich et al., Science 2018)
  - ~2% of a target population in a consumer database is enough
  - A third-cousin-or-closer match is then available for ~60% of people of that ancestry
- The 2026 twist: DNA embeddings from foundation models can be inverted to recover the sequence
  - An embedding is not de-identified data
- One of the five cases Week 14 re-opens. Start thinking now.



*Schematic, after Erlich et al. 2018 — toy model fitted to the 2% → 60% headline only*

# Design Notebook 1 — the five-step mould

- The five-step mould, used twice this semester:
  - DESIGN — given a seeded scenario and a hard budget, specify the experiment and justify two parameters
  - RUN — execute it on your own seed
  - CONTROL — run the comparison that would have revealed your design as inadequate
  - DEFEND — 120 words on what your result licenses you to claim, and what it does not (at home)
  - AUDIT — find, classify and fix the defect in supplied AI-written code (at home)



*The five-step mould — reuse unchanged for Design Notebook 2 in Week 9*

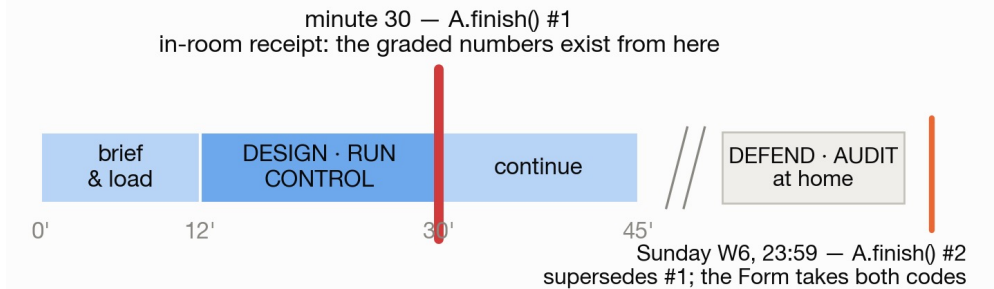
# Design Notebook 1 — deadline, budget, tier

- Due Sunday of WEEK 6, 23:59, EgeDers. Not this Sunday. You get eleven days.
- Budget it at ~2.5 hours. If it takes you five, tell me — that is a design fault, not your fault.
- Tier: BLUE. You are required to use an AI model, because judging its output is the assessed skill.

# The BLUE tier, and the hour ahead

- BLUE = AI use is REQUIRED and the interaction record is part of what you submit. (MIT Appendix B.3.)
- Why: evaluating, repairing and taking responsibility for model output IS the professional skill being assessed here
- Your ### AI DISCLOSURE cell must name at least one thing the model got wrong or unsuited, and how you detected it
- A submission reporting zero corrections is not evidence of a perfect model. It is evidence that nobody checked, and it is marked down on the AUDIT band.
- This afternoon: DESIGN, RUN, CONTROL in the room. A.finish() at minute 30. DEFEND and AUDIT at home.
- This is the notebook every micro-viva in Weeks 11–14 opens. For every one of you.

The hour ahead — and the eleven days after it



Nobody leaves the room without the graded payload existing.

*Minute 30: A.finish() #1; Sunday W6: A.finish() #2 supersedes it*

## APPENDIX (not taught in the 75 minutes — posted to EgeDers)

- Assembly: overlap → contigs → scaffolds → consensus. BIB Lecture\_06 slides 22–32.
- Illumina chemistry in detail: bridge amplification, cluster generation, SBS cycle. BIB Lecture\_06 slides 6–9, plus one linked video.
- Repeat classes and the human genome composition. BIB Lecture\_06 slides 14–21.
- Lander–Waterman:  $\text{Prob}[\text{base not covered}] = e^{(-C)}$ , under a uniformity assumption that does not hold.
- Mate-pair libraries. Historical; superseded by long reads.
- None of this is examinable. It is here because some of you will need it in an MSc.